

A Practical Guide to Lasso regression

TOPIC -- LASSO REGRESSION

$$L(\beta) = \|y - X\beta\|^2 + \lambda \|\beta\|_1$$

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where $\tilde{y} = y - X\beta_0$ $\{\displaystyle \tilde{y} = y - X\beta_0\}$. If $b = b_{\text{OLS}}$ $\{\displaystyle b = b_{\text{OLS}}\}$ when R^2 $\{\displaystyle R^2\}$ is computed, then the diagonal elements of $R \otimes$ $\{\displaystyle R^{\otimes}\}$ sum to R^2 $\{\displaystyle R^2\}$. The diagonal $R \otimes$ $\{\displaystyle R^{\otimes}\}$ values may be smaller than 0 or, less often, larger than 1. If regressors are uncorrelated, then the i^{th} $\{\displaystyle i^{\text{th}}\}$ diagonal element of $R \otimes$ $\{\displaystyle R^{\otimes}\}$ simply corresponds to the r^2 $\{\displaystyle r^2\}$ value between x_i $\{\displaystyle x_i\}$ and y $\{\displaystyle y\}$. A rescaled version of the adaptive lasso of can be obtained by setting $q_{\text{adaptive lasso}, i} = |b_{\text{OLS}, i} - \beta_0|$ $\{\displaystyle q_{\text{adaptive lasso}, i} = |b_{\text{OLS}, i} - \beta_0|\}$. If regressors are uncorrelated, the moment that the i^{th} $\{\displaystyle i^{\text{th}}\}$ parameter is activated is given by the i^{th} $\{\displaystyle i^{\text{th}}\}$ diagonal element of $R \otimes$ $\{\displaystyle R^{\otimes}\}$. Assuming for convenience that β_0 $\{\displaystyle \beta_0\}$ is a vector of zeros,

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where q_i $\{\displaystyle q_i\}$ is specified below and the "prime" symbol stands for transpose. The first fraction represents relative accuracy, the second fraction relative simplicity, and λ $\{\displaystyle \lambda\}$ balances between the two.

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Ensemble lasso Lasso can be run in an ensemble. This can be especially useful when the data is high-dimensional. The procedure involves running lasso on each of several random subsets of the data and collating the results.

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$$\min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{N} \|y - X\beta\|^2 + \lambda \|\beta\|_1 \right\} \quad \{\displaystyle \min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{N} \|y - X\beta\|^2 + \lambda \|\beta\|_1 \right\}\}$$

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$$\left\| \hat{\beta}_j - \arg\min_{\beta} \left(\sum_{i=1}^N \left(y_i - \beta_j^T x_i \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right) \right\|$$

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Least squares Consider a sample consisting of N cases, each of which consists of p covariates and a single outcome. Let y_i be the outcome and $x_i := (x_{i1}, x_{i2}, \dots, x_{ip})^T$ be the covariate vector for the i th case. Then the objective of lasso is to solve:

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In 2005, Zou and Hastie introduced the elastic net. When $p > n$ (the number of covariates is greater than the sample size) lasso can select only n covariates (even when more are associated with the outcome) and it tends to select one covariate from any set of highly correlated covariates. Additionally, even when $n > p$, ridge regression tends to perform better given strongly correlated covariates. The elastic net extends lasso by adding an additional ℓ_2 penalty term giving

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where $\|z\|_0$ is the " ℓ_0 norm", which is defined as $\|z\|_0 = m$ if exactly m components of z are nonzero. Again assuming orthonormal covariates, it can be shown that in this special case

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Like b_{ℓ_1} , b_{ℓ_2} moves in the direction of the point $(\lambda = R^2, b = 0)$ when λ is close to zero; but unlike b_{ℓ_1} , the influence of R^2 diminishes in b_{ℓ_2} if λ increases (see figure). Given multiple regressors, the moment that a parameter is activated (i.e. allowed to deviate from β_0) is also determined by a regressor's contribution to R^2 accuracy. First,

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$$\min_{\beta \in \mathbb{R}^p} \{ \|y - X\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2 \}, \quad \{\displaystyle \min_{\beta \in \mathbb{R}^p} \left\{ \left\| y - X\beta \right\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2 \right\}, \}$$

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$$\min_{\beta \in \mathbb{R}^p} \{ \|y - X\beta\|_2^2 \} \text{ subject to } \|\beta\|_1 \leq t. \quad \{\displaystyle \min_{\beta \in \mathbb{R}^p} \left\{ \left\| y - X\beta \right\|_2^2 \right\} \text{ subject to } \|\beta\|_1 \leq t. \}$$

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but with respect to different constraints: $\|\beta\|_1 \leq t$ for lasso and $\|\beta\|_2^2 \leq t$ for ridge. The figure shows that the constraint region defined by the ℓ_1 norm is a square rotated so that its corners lie on the axes (in general a cross-polytope), while the region defined by the ℓ_2 norm is a circle (in general an n-sphere), which is rotationally invariant and, therefore, has no corners. As seen in the figure, a convex object that lies tangent to the boundary, such as the line shown, is likely to encounter a corner (or a higher-dimensional equivalent) of a hypercube, for which some components of β are identically zero, while in the case of an n-sphere, the points on the boundary for which some of the components of β are zero are not distinguished from the others and the convex object is no more likely to contact a point at which some components of β are zero than one for which none of them are.

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Just as ridge regression can be interpreted as linear regression for which the coefficients have been assigned normal prior distributions, lasso can be interpreted as linear regression for which the coefficients have Laplace prior distributions. The Laplace distribution is sharply peaked at zero (its first derivative is discontinuous at zero) and it concentrates its probability mass closer to zero than does the normal distribution. This provides an alternative explanation of why lasso tends to set some coefficients to zero, while ridge regression does not.